Instagram Performance Analytics — Project Flow & Insights

Sugar Cosmetics · Data Analytics & AI portfolio project · Data window 19 Apr 2024 – 19 Apr 2025

Headline. Sugar Cosmetics' Instagram content performs consistently: 18.54% average engagement rate across 50,000 records. But none of the seven levers tested — post format, weekday, campaign, audience emotion, engagement source, hashtag and hashtag count — is a proven winner: best-to-worst gaps are only 0.5–1.1%, far below a 5% materiality line. The real levers are reach, deeper engagement (comments, saves, shares) and profile-to-follow conversion.

1. Business objective and questions

The marketing team publishes steadily but cannot tell which content, timing and hashtags to invest in. This project answers five business questions from the data:

- Which post type earns the most likes, comments, saves and shares?
- Which weekdays and months are associated with stronger performance?
- Which hashtags are linked to better engagement or reach?
- Are people seeing posts but not interacting?
- What should the team publish — and test — next?

How the dashboard answers them

The Streamlit dashboard has ten tabs. A shared sidebar (date range, post type, campaign, engagement source, audience emotion, hashtag) filters every view.

Dashboard tab	Question it answers
Executive Summary	How is the account performing overall? (KPIs, key takeaways, engagement mix)
Trends	How consistent is publishing, and how is engagement moving? (monthly posts, engagement rate, content mix)
Content	Which formats and campaigns perform best? (averages, lift vs overall, campaign × format heatmap)
Timing	Does the day of the week matter? (weekday engagement with confidence intervals, volume, weekday × format)
Funnel & Growth	Are people seeing posts but not interacting — and do they follow? (funnel, reach-gap scatter)
Hashtags	Which hashtags and hashtag counts are linked to stronger performance?
Audience	Who engages, where do they come from, and do they follow? (emotion, source quality, follower growth, follow conversion)
Top Posts	Which records stand out on likes, engagement rate, reach, comments, shares, saves or follows?

Dashboard tab	Question it answers
Explorer	Can the team build its own view? (any metric × any dimension, two-way heatmap)
Strategy & AI	What should we do next? (lever scorecard, recommended actions, campaign ideas)

2. Data understanding and validation

Source: sugar_cosmetics_instagram_data (.xlsx and .csv — verified identical). One row = one post-level record.

Check	Result	Meaning
Rows × columns	50,000 × 14	10 text/category fields, 8 numeric metrics
Missing values	0	No imputation required
Exact duplicate rows	0	No de-duplication required
Negative numbers	0	All counts valid
Reach greater than impressions	0	Logical relationship holds
Zero values	Shares: 9, Follows: 1	Valid counts — kept as zero, not imputed
Date range	19 Apr 2024 – 19 Apr 2025	First and last months are partial
Categories	4 post types, 5 campaigns, 4 sources, 5 emotions, 5 hashtags	Spellings consistent

> **Data caveats.** The file has no caption or post ID, so records are labelled "#ID · Format · Campaign · Date". With about 137 records per day, a row is unlikely to be one literal Instagram post, and the flat distributions suggest synthetic or aggregated data. Treat the findings as a validated method and framework; re-run on the brand's real Instagram Insights export before acting.

3. Project flow — from raw file to recommendations

#	Stage	Tool	Output
1	Understand the business problem	Brief, stakeholder questions	Objectives, questions, KPI list
2	Understand the dataset	Python (pandas)	Data dictionary, quality report
3	Clean and validate	preprocess.py	Validated CSV + preprocessed Excel
4	Create calculated fields	preprocess.py	Total Engagement, Engagement Rate, Month, Day of Week
5	Build the 10 visualizations	Plotly in Streamlit	10 titled, insight-annotated views
6	Build the interactive dashboard	Streamlit sidebar filters	Post type / month / campaign / source / hashtag filters
7	Interpret and recommend	Analysis + statistical test	Insights doc, business summary, deck

4. KPI definitions

KPI	Formula	Note
Total Engagement	Likes + Comments + Shares + Saves	Project definition, not Instagram's official one
Engagement Rate (%)	Total Engagement ÷ Reach × 100	Per record; headline = average of records
Profile Visit Rate (%)	Profile Visits ÷ Reach × 100	Interest in the account
Follow Conversion (%)	Follows ÷ Profile Visits × 100	Exploratory — attribution rules unverified
Reach Rate (%)	Reach ÷ Impressions × 100	Share of views that are unique accounts

Summed reach counts the same account once per record, so it is not the number of unique people reached.

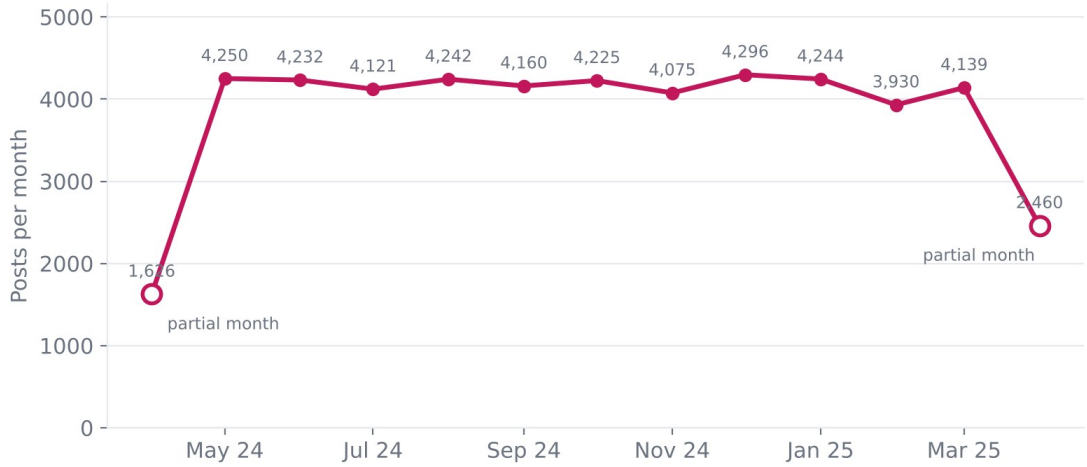
5. Findings by analysis

Task 1 — KPI Overview

50,000	147.73M	18.46M	987.42M	18.54%
Posts (records)	Likes	Comments	Total reach	Avg engagement rate

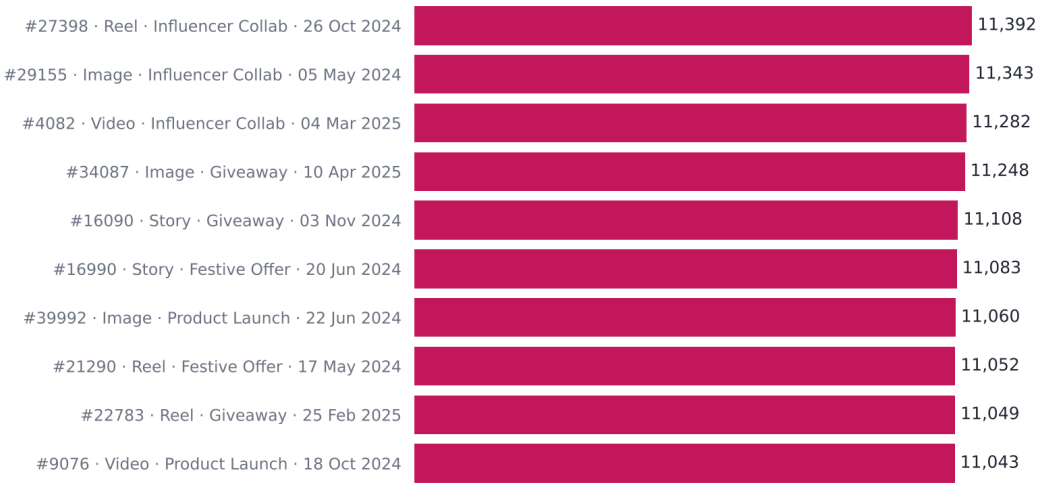
Total engagement is 183.16M on 1.28B impressions. Likes make up 81% of all interactions; comments 10%, saves 4.8%, shares 4.4%. Every 1,000 accounts reached produce about 185 interactions.

Task 2 — Monthly Posting Trend



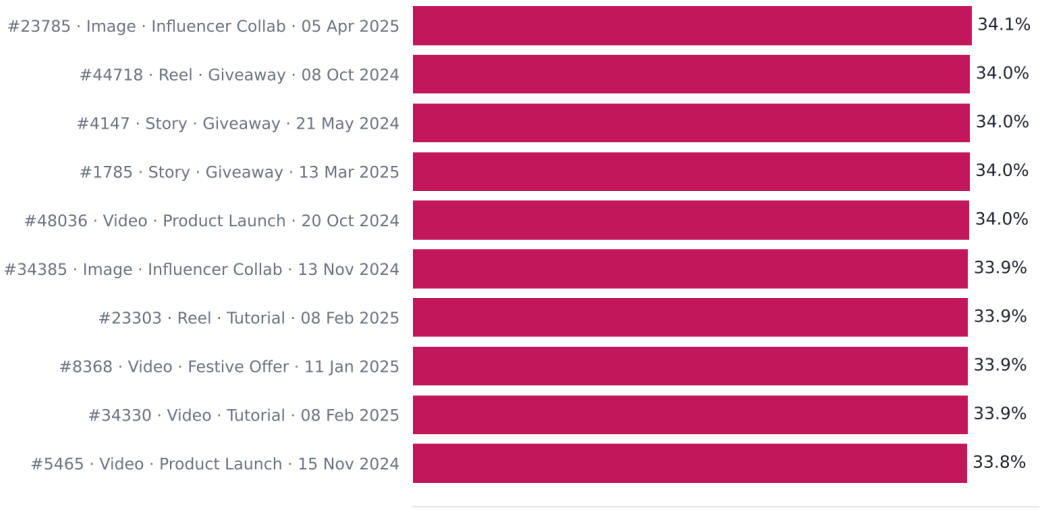
Publishing is steady at roughly 4,000–4,300 records a month. April 2024 and April 2025 look low only because the data covers just part of each month. Consistency is a strength; there is no seasonal posting gap.

Task 3 — Top 10 by Likes



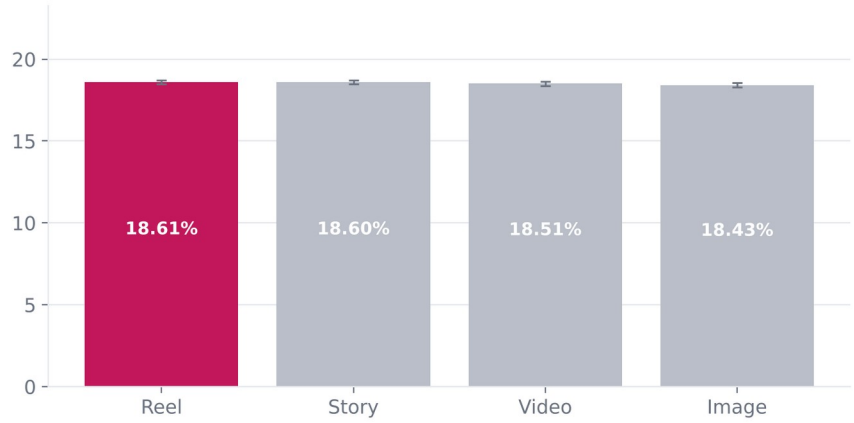
The top 10 average 45,308 reach against 19,748 overall. Likes and reach are strongly correlated (r = 0.79), so "most liked" largely means "most seen". Influencer Collab and Giveaway campaigns appear repeatedly.

Task 4 — Top 10 by Engagement Rate



These hidden gems reach 34.1% engagement against the 18.5% average, with a median reach of 23,486 — below average. High engagement does not need huge reach, and none of them appear in the top-10-by-likes list.

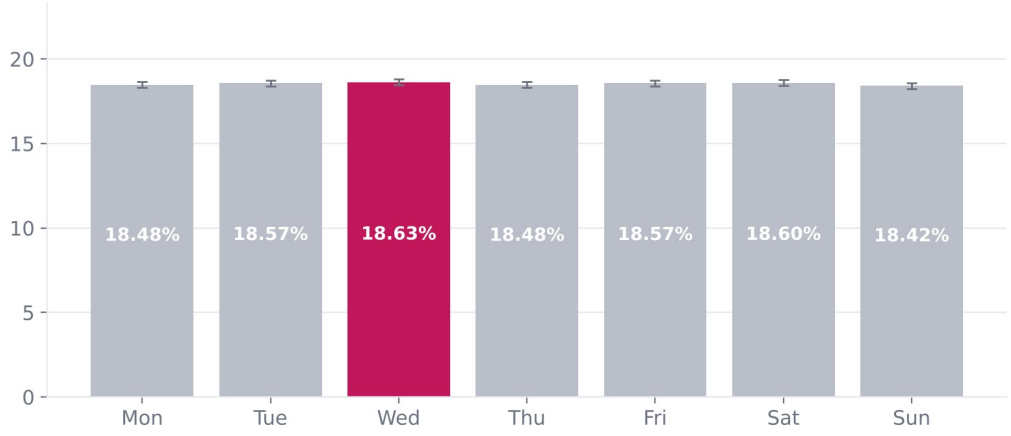
Task 5 — Post Type Performance



Post type	Records	Avg likes	Avg comments	Avg saves	Avg shares	Eng. rate
Reel	12,599	2,960	369	178	162	18.61%
Story	12,517	2,964	371	178	165	18.60%
Video	12,485	2,941	370	176	162	18.51%
Image	12,399	2,954	367	177	161	18.43%

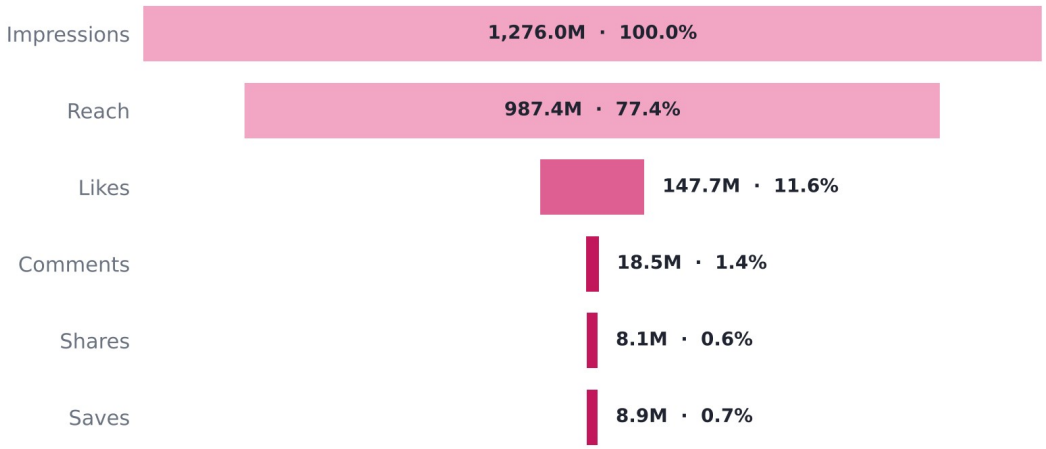
Reel leads on engagement rate (18.61%), but the best-to-worst gap is only 1.0% and the difference is not statistically significant (ANOVA p = 0.15). Format alone is not what drives results.

Task 6 — Day of Week



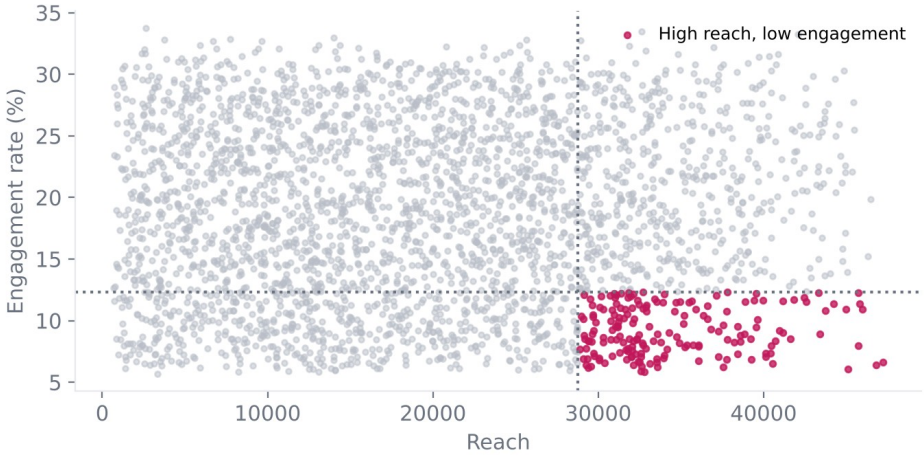
Wednesday is highest (18.63%) and Sunday lowest (18.42%) — a 0.21-point gap. Confidence whiskers overlap (p = 0.57), so there is no proven best day. Hour of posting is not in the data and should be captured.

Task 7 — Engagement Funnel



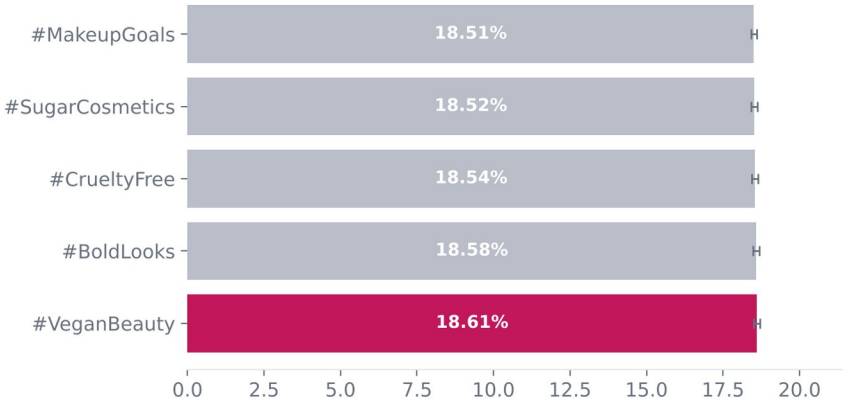
77 of every 100 impressions are unique accounts and about 12 become likes. Comments, shares and saves each convert under 1.5% of impressions — the big drop-off is from seeing to deeper actions. Of reached accounts, 3.0% visit the profile and 30% of visitors follow.

Reach without interaction



3,117 records (6.2%) combine top-quartile reach (≥ 28,763) with bottom-quartile engagement (≤ 12.3%). Overall, engagement rate is uncorrelated with reach (r = 0.00): more reach brings more interactions, not better ones. These records are the clearest optimisation target.

Task 8 — Hashtag Analysis

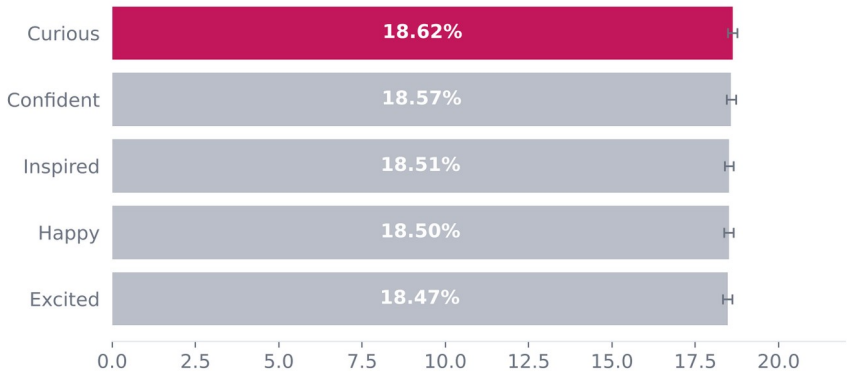


#VeganBeauty leads (18.61%) but all five hashtags sit within 0.10 points of each other and their confidence ranges overlap. Posts with 2 hashtags score highest (18.63% vs 18.44% for 1 hashtag). That gap is the largest hashtag-related signal and sits just outside conventional significance (ANOVA p = 0.054), so it is the first thing worth A/B testing.

Content lift versus the overall average

Relative to the 18.54% account average, Reel is +0.4% and Story +0.34%, while Image is -0.59% and Video -0.16%. By campaign, Influencer Collab is +0.32%, Tutorial +0.26%, Giveaway +0.16%, Product Launch -0.22% and Festive Offer -0.53%. Every lift is well under 1% — directionally useful, but too small to justify shifting budget on its own.

Audience — who engages and where from

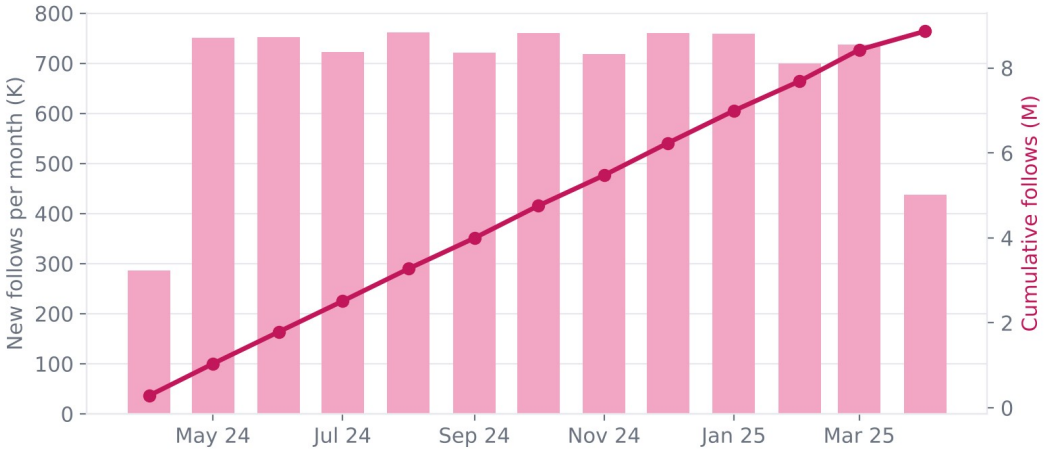


Curious content has the highest engagement rate (18.62%) and Excited the lowest (18.47%); whiskers overlap (p = 0.56), so emotion is not a reliable lever.

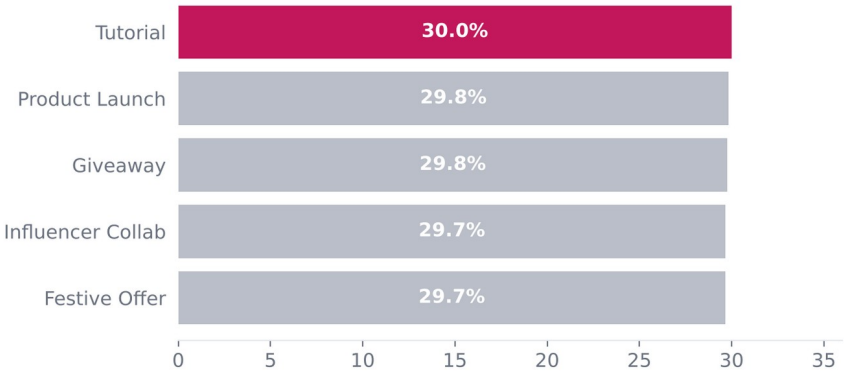
Engagement source	Records	Avg reach	Avg engagement rate	Avg follows
Collab	12,633	19,707	18.59%	177
Influencer	12,501	19,699	18.58%	175
Paid	12,358	19,617	18.53%	178
Organic	12,508	19,970	18.44%	179

Source quality is nearly identical. Organic posts reach the most accounts on average (19,970) but convert slightly less of that reach into interactions (18.44%); Collab leads on engagement rate (18.59%). The spread is 0.8% (p = 0.35), so paid, influencer, collab and organic are performing alike.

Follower growth and conversion



The account gained 8.87M follows over the period — about 740,236 in a typical full month — with growth tracking posting volume. 3.0% of reach visits the profile and 29.9% of visitors follow (follows ÷ profile visits; exploratory because attribution rules are unverified).

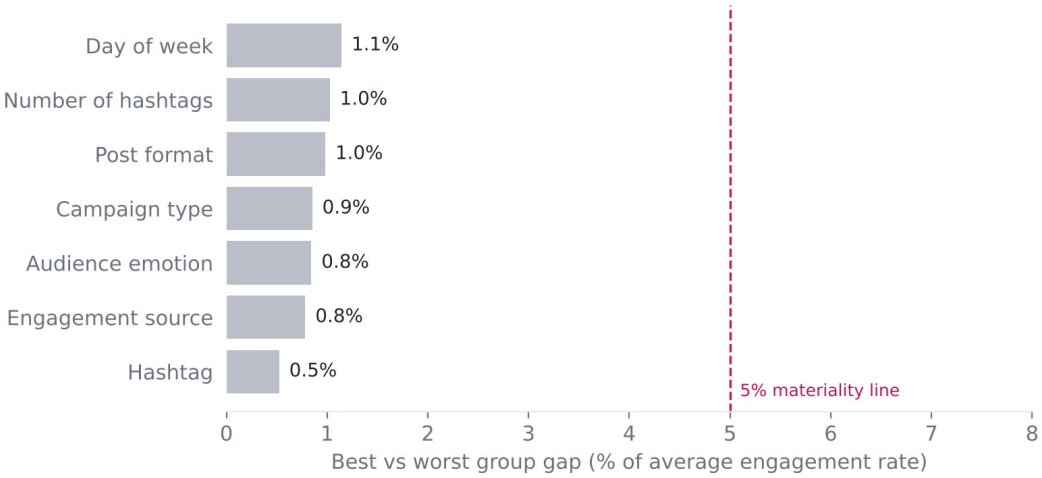


Average profile-visit-to-follow conversion by campaign is almost identical (about 29.7–30.0%): Tutorial is highest, Festive Offer and Influencer Collab lowest. Conversion is therefore a profile-level lever (bio, highlights, link-in-bio), not a campaign-level one.

Top posts across every metric

The Top Posts tab ranks records by likes, engagement rate, reach, comments, shares, saves or follows. Whatever the metric, the leaders are large-reach records: the top 8 for each metric average 43,848–45,553 reach against 19,748 overall. Reach correlates strongly with likes (r = 0.79), comments (0.68), saves (0.66), follows (0.66) and shares (0.62) — but not with engagement rate (0.00). The only ranking that surfaces smaller, over-performing records is engagement rate.

Lever scorecard — are any of these differences real?



Lever	Best group	Worst group	Gap	ANOVA p	Verdict
Day of week	Wednesday	Sunday	1.1%	0.57	Within noise
Number of hashtags	2 hashtags	1 hashtag	1.0%	0.05	Borderline — test
Post format	Reel	Image	1.0%	0.15	Within noise
Campaign type	Influencer Collab	Festive Offer	0.9%	0.48	Within noise
Audience emotion	Curious	Excited	0.8%	0.56	Within noise

Lever	Best group	Worst group	Gap	ANOVA p	Verdict
Engagement source	Collab	Organic	0.8%	0.35	Within noise
Hashtag	#VeganBeauty	#MakeupGoals	0.5%	n/a	Within noise (CIs overlap)

Gap = (best group − worst group) ÷ average. Hashtags are compared on non-overlapping 95% confidence intervals because a post can carry several hashtags.

No lever clears the bar of a gap of at least 5% with statistical support. Only the number of hashtags comes close to significance (p = 0.054), which is why a 2-vs-1/3 hashtag test is the first experiment recommended.

6. Strategic recommendations

Action	What to do	Evidence
Scale reach	Invest in amplification: influencer collabs, boosted proven creatives, Reels/Stories distribution.	Engagement rate is flat as reach grows; ~185 interactions per 1,000 reached.
Fix 'seen but ignored'	Audit high-reach, low-engagement posts for weak hooks and missing calls-to-action; add polls, questions, giveaways.	6.2% of records (3,117).
Deepen engagement	Create save-worthy tutorials and share-worthy festive/challenge content.	Likes are 81% of interactions.
Convert visitors	Strengthen bio, highlights, pinned posts and link-in-bio offers; add 'follow for shade drops' end frames.	3.0% visit profile; 30% of visitors follow, regardless of campaign.
Test, don't guess	Run 4-week A/B tests: 2 vs 1/3 hashtags first, then Reel vs Image and morning vs evening. Act only on significant results.	All seven lever gaps are 0.5–1.1%; hashtag count is borderline (p = 0.05).
Keep the base	Maintain steady cadence, balanced format mix and the five brand/value hashtags.	Posting volume and engagement are consistent month to month.
Close data gaps	Capture post ID, caption, posting hour, carousel format, ad spend and follower count.	Needed for timing, caption, ROI and growth analysis.

Content and campaign ideas

- Tutorial Reels series ("5-minute festive look") — built for saves and shares.
- Influencer collab takeovers with a giveaway mechanic — built for reach plus comments.
- Festive Offer Stories with poll and quiz stickers — interaction at scale.
- Vegan and cruelty-free brand-values content — on-brand, consistently performing hashtag themes.
- Profile-visit nudges in the last frame of high-reach posts.